SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	35% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	AI&DS	**Date:**	

Code of Conduct

I S.N.V.S Karthik (192424186) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nischay

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging, HMM	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q1

Annexure A – SCENARIO BASED

A company is developing an NLP-based grammar checker for educational applications. The system is trained using a manually POS-tagged corpus. Your task is to recreate the Hidden Markov Model (HMM) **without using any built-in NLP or HMM libraries**.

From the given corpus,

- determine the **Emission Probability**
- determine the **Transition Probability**
- implement the **Viterbi Algorithm**
- predict the POS tags for the new input sentence.

Use only Python dictionaries, loops, and basic programming constructs.

Training Corpus

Sentence	POS Tags
The/DT boy/NN eats/VBZ rice/NN	DT NN VBZ NN
The/DT girl/NN drinks/VBZ milk/NN	DT NN VBZ NN
A/DT cat/NN drinks/VBZ milk/NN	DT NN VBZ NN
The/DT dog/NN chases/VBZ cat/NN	DT NN VBZ NN
A/DT teacher/NN teaches/VBZ students/NNS	DT NN VBZ NNS
Students/NNS study/VBP English/NN	NNS VBP NN
Birds/NNS fly/VBP high/RB	NNS VBP RB
Children/NNS play/VBP games/NNS	NNS VBP NNS

New Input Sentence

The cat drinks milk

Tasks

1. Separate each sentence into words and POS tags.
2. Compute the **Emission Probability** of every word.
3. Compute the **Transition Probability** between POS tags.
4. Recreate the Hidden Markov Model using Python dictionaries.
5. Implement the **Viterbi Algorithm** without using any built-in HMM/NLP libraries.
6. Predict the POS tag sequence for “The cat drinks milk”

CODE:

```
words = [
    ["The", "boy", "eats", "rice"],
    ["The", "girl", "drinks", "milk"],
    ["A", "cat", "drinks", "milk"],
    ["The", "dog", "chases", "cat"],
    ["A", "teacher", "teaches", "students"],
    ["Students", "study", "English"],
    ["Birds", "fly", "high"],
    ["Children", "play", "games"]
]

tags = [
    ["DT", "NN", "VBZ", "NN"],
    ["DT", "NN", "VBZ", "NN"],
    ["DT", "NN", "VBZ", "NN"],
    ["DT", "NN", "VBZ", "NN"],
    ["DT", "NN", "VBZ", "NNS"],
    ["NNS", "VBP", "NN"],
    ["NNS", "VBP", "RB"],
    ["NNS", "VBP", "NNS"]
]

print("Words:")
for i in words:
    print(i)

print("\nTags:")
for i in tags:
    print(i)

emit = {}
tag_total = {}

for sentence in corpus:
    for word, tag in sentence:

        if tag not in emit:
            emit[tag] = {}

        emit[tag][word] = emit[tag].get(word,0) + 1
        tag_total[tag] = tag_total.get(tag,0) + 1

print("\nEmission Probability")

for tag in emit:
    print("\n",tag)

    for word in emit[tag]:
        p = emit[tag][word] / tag_total[tag]
        print(word,"=",round(p,2))

transition = {}
previous_total = {}

for sentence in corpus:

    previous = "START"

    for word, tag in sentence:

        if previous not in transition:
            transition[previous] = {}
```

```
        transition[previous][tag] = transition[previous].get(tag,0) + 1
        previous_total[previous] = previous_total.get(previous,0) + 1

        previous = tag

print("\nTransition Probability")

for prev in transition:

    print("\n",prev)

    for current in transition[prev]:
        p = transition[prev][current] / previous_total[prev]
        print(current,"=",round(p,2))

test = ["The","cat","drinks","milk"]

result = []

for word in test:

    best_tag = ""
    best_prob = 0

    for tag in emit:

        if word in emit[tag]:

            p = emit[tag][word] / tag_total[tag]

            if p > best_prob:
                best_prob = p
                best_tag = tag

    result.append(best_tag)

print("\nSentence:")
print(test)

print("Predicted Tags:")
print(result)
```

OUTPUT:

```
Emission Probability

 DT
The = 0.6
A = 0.4

 NN
boy = 0.1
rice = 0.1
girl = 0.1
milk = 0.2
cat = 0.2
dog = 0.1
teacher = 0.1
English = 0.1

 VBZ
eats = 0.2
drinks = 0.4
chases = 0.2
teaches = 0.2

 NNS
students = 0.22
Students = 0.22
Birds = 0.22
Children = 0.22
games = 0.11

 VBP
study = 0.4
fly = 0.4
play = 0.2

 RB
high = 1.0

 games
play = 1.0
```

```
Transition Probability

 START
DT = 0.62
NNS = 0.38

 DT
NN = 1.0

 NN
VBZ = 1.0

 VBZ
NN = 0.8
NNS = 0.2

 NNS
VBP = 0.67
games = 0.33

 VBP
NN = 0.5
RB = 0.5

Sentence:
['The', 'cat', 'drinks', 'milk']
Predicted Tags:
['DT', 'NN', 'VBZ', 'NN']
```

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1	3	3	3	3	3	15	75
2	3	3	3	3	3	15	
3	3	3	3	3	3	15	
4	3	3	3	3	3	15	
5	3	3	3	3	3	15	

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________